
HAOLIN YANG

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EDUCATION

University of Chicago

Sept. 2024 - Now

Master of Science in Statistics (GPA: 4.0/4.0)

Courseworks: Introduction to Computer Vision, Stochastic Calculus, Learning, Decisions, and Limits

Tsinghua University

Sept. 2020 - June 2024

Bachelor of Arts in English **Outstanding Graduate** (GPA: 3.94/4.0)

Minor in Economics and Finance (GPA: 4.0/4.0)

Minor in Statistics (without degree) (GPA: 4.0/4.0)

Courseworks: Nonparametric Statistics, Linear Regression Analysis, Real Analysis, Econometrics, Applied Stochastic Processes

HONOR AND AWARDS

Academic Excellence Scholarship (2022-2023)

Outstanding Graduate of the School of Humanities, Tsinghua University (2024)

Outstanding Undergraduate Thesis of the School of Humanities, Tsinghua University (2024)

RESEARCH INTERESTS

Large Language Models, In-context Learning, Interpretability

PUBLICATIONS AND MANUSCRIPTS

1 Zhang, Y., Yang, F., Yang, H., & Han, S. (2024). Does checking-in help? Understanding L2 learners' autonomous check-in behavior in an English-language MOOC through learning analytics. *ReCALL*, 1–16. [doi:10.1017/S0958344024000144](https://doi.org/10.1017/S0958344024000144)

RESEARCH EXPERIENCES

Demonstration Selection for In-context learning From a Bayesian Perspective

Oct. 2024 - Now

Advisor: Kaize Ding

- Unite the Bayesian interpretation of ICL and empirical findings in demonstration selection by proposing a multivariate Bayesian characterization of ICL based on LLM hidden states
- Devise methods to improve the ICL performance that can be integrated into any framework of demonstration selection
- Prove that LLM layer updates implicitly performs Bayesian inference

In-context learning as semantic clustering plus fuzzy copy

July. 2024 - Now

Advisor: Yiqiao Zhong

- Discovered patterns of semantic clustering in the early layer hidden representations of different in-context demonstrations inside LLMs by investigating the (non)linear separability of the representations at different layers
- Provide empirical evidence that LLMs is approximately performing fuzzy copy among the hidden representations of icl prompts by matching the embedding matrices of different attention heads of LLMs
- Propose a two-stage characterization of the ICL mechanisms and validate its existence in various form of NLP and reasoning tasks

Undergraduate Thesis:

Automatic Post-Editing of English-Chinese Machine-Translated Academic Texts Jan. 2024 - June 2024

- Designed and fine-tuned an automatic post-editing model and tested its performance on ParaMed, a English-Chinese parallel corpus that contains academic texts
- Performed comparative statistical analyses as concerning the BLEU and TER scores of the raw machine-translated texts and the automatically post-edited texts
- Qualitatively examined the performance of the model by investigating how it helped to correct the machine translation errors and its failure scenarios

Factor Analysis and Language Education Assessment Instrument

Feb. 2023 – June 2023

- Wrote two chapters on factor analysis in a monograph titled *Statistical Methods in Language Education Studies*, which is currently in press
- Analyzed the similarities and differences between principal component analysis and exploratory factor analysis in theoretical and practical terms and discussed how they tend to be misused in many language education studies
- Applied the two analytical methods to the data collected by administering the Chinese Version of the Foreign Language Enjoyment Scale (C-FLES) to 260 high school English learners in China
- Used the two analytical methods to confirm a three-factor latent construct that underlies the 11 items in the scale and identified two items as being potentially redundant

Check-in Behaviors and Academic Performance in MOOC

March 2021 – April 2022

- Collected and cleaned the academic performance data of 11296 students of an English MOOC on the Xue-TangX platform including their check-in instances and scores in assignments and tests
- Conducted chi-square test of independence and logistic regression to reveal to what extent students' check-in behaviors affect their course completion rates
- Conducted Mann-Whitney U tests between the students who checked in regularly and those who did not check in to determine the differences in their learning behaviors

SKILLS

Statistics / Data Analysis:

Proficiency in R, Python, Stata

Languages:

TOEFL 117: Reading 30, Listening 30, Writing 30, Speaking 27;

GRE 339+5.0: Verbal 169, Quantitative 170, AW 5.0;

Mandarin (native), German (C1)

Graduate

Academic Program History

Program: Physical Sciences Division
Start Quarter: Autumn 2024
Current Status: Active in Program
M.S. in Statistics

External Education

Tsinghua University
Beijing, China
Bachelor of Arts 2024

Tsinghua University
Beijing, China
Bachelor of Economics 2024

Beginning of Graduate Record

Autumn 2024

<u>Course</u>	<u>Description</u>	<u>Attempted</u>	<u>Earned</u>	<u>Grade</u>
STAT 30030	Statistical Theory and Methods Ia	100	100	A
STAT 37711	Foundations of Machine Learning and AI - Part I	100	100	A
TTIC 31040	Introduction to Computer Vision	100	100	A

End of Graduate

TSINGHUA UNIVERSITY

ACADEMIC TRANSCRIPT

Student Name Yang Haolin

Gender Male **Student No.** 2020011886 **Student Type** Undergraduate **Date of Admission** September,2020

School/Department Department of Foreign Languages and Literatures **Major** English Language

Course Number	Course Title	Credit	Grade	Point	Year-Semester
10421263	Calculus C(1)(For Social Science)	3	B+	3.6	2020-Autumn
10421284	Linear Algebra (For Social Science)	4	A+	4.0	2020-Autumn
10610183	Ideological Moral and Legal Education	3	A-	4.0	2020-Autumn
10691342	Writing and Communication	2	A	4.0	2020-Autumn
10720011	Physical Education(1)	1	C+	2.6	2020-Autumn
12090052	Military Theory	2	B+	3.6	2020-Autumn
14000052	Medical Psychology and Health Life	2	A	4.0	2020-Autumn
20740073	Programming Fundamentals	3	A-	4.0	2020-Autumn
30642074	English Writing and Critical Thinking (1)	4	A	4.0	2020-Autumn
30642083	Reading and Critical Thinking (1)	3	A	4.0	2020-Autumn
10691412	Confucius and His Critics	2	W	N/A	2020-Autumn
01510492	Energy and Society	2	A-	4.0	2021-Spring
02090011	Introduction to Sun Wu's Art of War	1	A	4.0	2021-Spring
10610193	Outline of Modern Chinese History	3	B+	3.6	2021-Spring
10720021	Physical Education(2)	1	C+	2.6	2021-Spring
30642093	Reading and Critical Thinking (2)	3	A	4.0	2021-Spring
30700653	The Principles of Economics	3	A	4.0	2021-Spring
40641423	Introduction to Linguistics (1)	3	A-	4.0	2021-Spring
40641853	History of English Literature	3	A-	4.0	2021-Spring
40641862	Classical Latin (1)	2	A-	4.0	2021-Spring
10420803	Probability and Statistics	3	W	N/A	2021-Spring
12090062	Military Skills	2	A-	4.0	2021-Summer
40642003	Acting Shakespeare	3	P	N/A	2021-Summer
00782082	Introduction to Film and TV Arts and works Appreciation	2	A	4.0	2021-Autumn
10420803	Probability and Statistics	3	B	3.3	2021-Autumn
10610204	Principle of Marxist Philosophy	4	A-	4.0	2021-Autumn
10720031	Physical Education(3)	1	B-	3.0	2021-Autumn
14205094	Russian (Second Foreign Language) (1)	4	A	4.0	2021-Autumn
30510123	Accounting Principles	3	A+	4.0	2021-Autumn
30510743	Intermediate Microeconomics	3	A	4.0	2021-Autumn
30510833	Principles of Economics(1)	3	A	4.0	2021-Autumn
40640772	the History of the English Language	2	A-	4.0	2021-Autumn
40641242	E-C & C-E Translation (1)	2	A	4.0	2021-Autumn
40641673	English Poetry	3	A	4.0	2021-Autumn
10421315	Calculus A(2)	5	A	4.0	2022-Spring
10680022	Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	2	A-	4.0	2022-Spring
10680032	Introduction to MaoZedong Thought and the Theoretical System of Socialism with Chinese Characterist1	2	A-	4.0	2022-Spring
10700163	Understanding Psychology	3	B+	3.6	2022-Spring
10720041	Physical Education(4)	1	A	4.0	2022-Spring
30510803	Principles of Economics(2)	3	A+	4.0	2022-Spring
30510973	Econometrics(1)	3	A	4.0	2022-Spring
30511053	Corporate Finance	3	A	4.0	2022-Spring
40640812	Introduction to Syntax	2	A	4.0	2022-Spring

Total Credits: 175.0 **GPA:** 3.94

Date of Graduation: *****

Degree Conferred: *****

Director of Registrar's Office:

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Official Seal:

Date Printed: October 10, 2023

TSINGHUA UNIVERSITY

ACADEMIC TRANSCRIPT

Student Name Yang Haolin

Gender Male **Student No.** 2020011886 **Student Type** Undergraduate **Date of Admission** September,2020

School/Department Department of Foreign Languages and Literatures **Major** English Language

Course Number	Course Title	Credit	Grade	Point	Year-Semester
40641262	E-C & C-E Translation (2)	2	A-	4.0	2022-Spring
40641603	Reading Shakespeare	3	P	N/A	2022-Spring
42540012	Students Research Training	2	A	4.0	2022-Spring
10680042	Introduction to MaoZedong Thought and the Theoretical System of Socialism with Chinese Characterist2	2	P	N/A	2022-Summer
10680011	Situation and Policy	1	A-	4.0	2022-Autumn
10720110	Physical Education(1)	0	C-	2.0	2022-Autumn
30641572	Interpreting(1)	2	A-	4.0	2022-Autumn
30642113	Introduction to Psychology of Reading	3	A+	4.0	2022-Autumn
40160753	Applied Stochastic Processes	3	P	N/A	2022-Autumn
40510943	Theory of Industrial Organization	3	A	4.0	2022-Autumn
40511003	Environmental and Resource Economics	3	A	4.0	2022-Autumn
40512103	Health Economics	3	A+	4.0	2022-Autumn
40640742	An Introduction to Semantics	2	A-	4.0	2022-Autumn
40641593	History of Western Linguistics	3	P	N/A	2022-Autumn
00690022	Sino-European Relations in Ancient Time	2	W	N/A	2022-Autumn
00310332	Creative Design in Mechanics	2	P	N/A	2023-Spring
00692342	Science Museum Studies	2	A-	4.0	2023-Spring
10510064	Real Analysis	4	A-	4.0	2023-Spring
10691543	Tsinghua Bamboo Slips and Ancient Civilization	3	A	4.0	2023-Spring
10720120	Physical Education(2)	0	C+	2.6	2023-Spring
30510763	Intermediate Macroeconomics	3	A	4.0	2023-Spring
30640132	English Listening(2)	2	A+	4.0	2023-Spring
30641162	English Speech	2	A+	4.0	2023-Spring
40160803	Linear Regression Analysis	3	A	4.0	2023-Spring
40640682	English Interpretation	2	A	4.0	2023-Spring
40641332	Literary Translation	2	A	4.0	2023-Spring
40642023	Language as Evidence	3	A	4.0	2023-Spring
40642102	English Thematic Reading (2)	2	A-	4.0	2023-Spring
40641973	Internship	3	A	4.0	2023-Summer
42540023	Students Research Training	3	A+	4.0	2023-Autumn

Total Credits: 175.0 **GPA:** 3.94

Date of Graduation: *****

Degree Conferred: *****

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KEY TO TRANSCRIPT

I. COURSE NUMBERING SYSTEM

Each course number consists of 8-10 characters.

The first character indicates the course level:

0-4 or H-T, W = undergraduate courses

6-9, A-G or X-Z = graduate courses

II. CREDIT

Credit is reported in terms of semester hours, whether earned during a 16-week semester or a summer session. For 1 unit of credit, either one hour per week is allotted to lecture or discussion, or two hours per week are allotted to laboratory, while more hours are needed for preparation or subsequent reading and study.

III. THE RECORD ENDS WITH *****.

IV. DATE OF GRADUATION and DEGREE CONFERRED

For currently enrolled undergraduates, the columns of DATE OF GRADUATION and DEGREE CONFERRED are *****.

V. GRADING SYSTEMS

a) EFFECTIVE for students who matriculated in spring 2015 and after

(i) Tsinghua University converted to a LETTER GRADING SYSTEM. The table below shows the grades in detail.

(ii) Credits are given for A+, A, A-, B+, B, B-, C+, C, C-, D+, D, P and EX.

(iii) W: Withdrew.

(iv) I: Incomplete. Marked when a student's application is approved for not attending the final exam.

(v) EX: Exemption. Students receive credits for exempted courses.

Grade	Grade Points	Corresponding 100-point Range	Equivalent 100-point value*
A+	4.0	95-100	100
A			98
A-			92
B+	3.6	85-89	87
B	3.3	80-84	82
B-	3.0	77-79	78
C+	2.6	73-76	75
C	2.3	70-72	71
C-	2.0	67-69	68
D+	1.6	63-66	65
D	1.3	60-62	61
F	0	0-59	0
P	N/A	N/A	N/A
F	N/A	N/A	N/A

* For the transition period in 2015-2018 between the 100-point grading system and the letter grading system, Tsinghua has provided a corresponding average of values in the 100-point range of each grade. The equivalent 100-point value for course receiving credits corresponds to the median in the range. Students who matriculated in spring 2019 and after no longer use the equivalent 100-point value.

b) EFFECTIVE for students who matriculated prior to spring 2015

(i) 100-POINT GRADING SYSTEM: Credits are given for 60 points and above.

(ii) PASS/FAIL SYSTEM: Credits are given for PASS.

DISTINCTION (for undergraduates only): Credits are given for DISTINCTION.

(iii) REPEATED COURSES: The transcript displays only the latest result of a repeated course. Repeated courses are designated with an "Rn" code beside the final grade, where "n" indicates the number of times the course was repeated.

VI. GRADING POLICY REFORM 2015-2018

In the ten years prior to spring 2015, 30 percent of A-range grades have been given. From fall 2015, Tsinghua initiated a grading reform: A-range grades (A+, A, A-) were to account for 20 percent of the grades given in all courses. In Spring 2019, the faculty reaffirmed its commitment to fair and transparent assessment and removed its numeric target for the percent of A-range grades.

VII. GPA CALCULATION

$$GPA = \frac{\sum \text{Course Credit} * \text{Grade Point}}{\sum \text{Course Credit}}$$

GPA is shown for students who matriculated in spring 2015 and after in a 4.0 grading scale. Course grades with N/A (Not Applicable) should not be included in GPA calculation.